

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414547
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Chefetz; Nathan et al.

Pet leash

Abstract

A twin pet leash prevents tangling by permitting the reel portion to rotate relative to the handle portion, and features individually controllable leash stop mechanisms. Even while an individual leash is stopped, the reel portion can rotate relative to the handle portion so that tangling of the pet leashes is avoided. The leash handle portion has two triggers that are color coded to match the color of the leash, so that a user can readily identify which of the two leashes are to be stopped. A ball bearing and/or a point contact reduces rotational friction on the reel portion while one or both of the brakes are activated.

Inventors:	Chefetz; Nathan (Kensington, MD), Goldenberg; Cade (Great Falls, VA)
Applicant:	Wigzi, LLC (Arlington, VA)
Family ID:	1000008684255
Assignee:	WIGZI LLC (Arlington, VA)
Appl. No.:	19/235300
Filed:	June 11, 2025

Publication Classification

Int. Cl.: A01K27/00 (20060101); B65H75/44 (20060101)

U.S. Cl.:

CPC A01K27/004 (20130101); B65H75/4442 (20130101);

Field of Classification Search

CPC: A01K (27/004); A01K (27/003); A01K (27/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6792893	12/2003	Quintero	119/796	A01K 27/004
7207296	12/2006	DiDonato	N/A	N/A
7455034	12/2007	DiDonato	N/A	N/A
7559292	12/2008	Blandford	N/A	N/A
7980202	12/2010	Bentz et al.	N/A	N/A
8256385	12/2011	Goldenberg	N/A	N/A
2011/0067649	12/2010	O'Brien	119/796	A01K 27/004
2011/0180017	12/2010	Goldenberg	119/796	A01K 27/004
2014/0238314	12/2013	O'Brien	119/796	A01K 27/004

Primary Examiner: Hayes; Kristen C

Attorney, Agent or Firm: Sterne, Kessler, Goldstein & Fox P.L.L.C.

Background/Summary

BACKGROUND

(1) This disclosure relates to the field of pet leashes. Specifically, this disclosure is related to a dual, retractable pet leash that prevents tangling by permitting the reel portion to rotate relative to the handle portion, and features individually controllable leash stop mechanisms

(2) Retractable pet leashes function by wrapping a leash around a reel that includes a mechanism that automatically rotates the reel to retract the leash. These leashes typically have a single reel with a single trigger mechanism for controlling the stopping of the reel. Single leash units having two leashes are known, for walking two pets at the same time. When pets pull on their leash, unwinding of a reel occurs; however sometimes the leash must be stopped from unwinding further, for example to prevent a pet from running into traffic or to prevent the pet from approaching another animal or human. Some leashes incorporate two reels allow a user to individually control each leash while including a rotation mechanism that reduces the potential for tangling. The disclosure below describes improvements to such dual leash reels, specifically regarding the rotation mechanism.

BRIEF SUMMARY

(3) In a first embodiment, a pet leash includes a handle configured to be held by a user; a first trigger disposed on the handle and positioned to be actuated by the user; and a second trigger disposed on the handle and positioned to be actuated by the user. A reel casing is rotatably connected to the handle, with a first reel disposed in the reel casing and having a first leash wrapped around the first reel and a second reel disposed in the reel casing and having a second leash wrapped around the second reel. A brake is rotatably disposed in the handle and extending into the reel casing, the brake including a first brake member slidably disposed in the brake such that the first brake member can slide into and out of contact with the first reel in the reel casing and a second brake member slidably disposed in the brake such that the second brake member can slide into and out of contact with the second reel in the reel casing. A first actuator arm is pivotably mounted in the handle and linked to the first trigger such that movement of the first trigger pivots the first actuator arm, wherein the first actuator arm is configured to pivot to displace the first brake

member towards the first reel. A ball bearing is disposed at an end of the first actuator arm and positioned to contact the first brake member when the first actuator arm pivots to displace the first actuator arm. A second actuator arm is pivotably mounted in the handle and linked to the second trigger such that movement of the second trigger pivots the second actuator arm, wherein the second actuator arm is configured to pivot to displace the second brake member towards the second reel. A contact member is disposed at an end of the second actuator arm and positioned to contact the second brake member when the second actuator arm pivots to displace the first actuator arm.

(4) In a further embodiment, the first actuator arm is configured to position the ball bearing such that an outer surface of the ball bearing is parallel to a surface of first brake member that is in contact with the ball bearing.

(5) In a further embodiment, the first actuator arm is configured to position the ball bearing such that an outer surface of the ball bearing is not parallel to a surface of first brake member that is in contact with the ball bearing.

(6) In a further embodiment, the ball bearing comprises a curved outer surface such that the contact between the first brake member and the ball bearing contacts the curved outer surface.

(7) In a further embodiment, a contact surface of the first brake member that contacts the ball bearing is an annular surface.

(8) In a further embodiment, the first actuator arm is configured to bring a rotational axis of the ball bearing parallel to a radial direction of the annular surface of the first brake member when the ball bearing contacts the annular surface.

(9) In a further embodiment, the contact member comprises a spherical element fixed to the end of the second actuator arm.

(10) In a further embodiment, the pet leash includes a wear element positioned on the second brake member at the location of contact between the second brake member and the contact member, wherein the wear element is configured to reduce wear of the second brake member caused by the contact member.

(11) In a further embodiment, the wear element comprises a metal disk that is fixed to a surface of the second brake member.

(12) In a further embodiment, contact member is configured to create a point contact between the second actuator arm and the second brake member.

(13) Certain aspects of the disclosure have other steps or elements in addition to or in place of those mentioned above. The steps or elements will become apparent to those skilled in the art from a reading of the following detailed description when taken with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS/FIGURES

- (1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles thereof and to enable a person skilled in the pertinent art to make and use the same.
- (2) FIG. 1 is a perspective view of a dual retractable pet leash with ability to stop pets individually, according to the present invention.
- (3) FIG. 2 is a perspective view from the side of the pet leash of FIG. 1.
- (4) FIG. 3 is a simplified schematic view showing the relative rotation of the two main parts of the leash of FIG. 1, with the two main parts being shown at a ninety degree angle to each other.
- (5) FIG. 4 is a perspective view of the leash of FIG. 1 showing the trigger elements and handle portion and showing the contours of the other parts.

- (6) FIG. 5 is an end view of the leash of FIG. 1, showing the trigger portion wherein the handle portion is uppermost and the reel portion is lowermost.
- (7) FIG. 6 is a perspective view of the interior portion of the leash of FIG. 1, showing the main operating parts and the reels.
- (8) FIG. 7 is a schematic perspective view showing only the reel portion and the portion of the stopping mechanism which engages directly with the reel portion, of the leash of FIG. 1.
- (9) FIG. 8 is a schematic perspective view of the reel portion and stopping mechanism of FIG. 7 as viewed from below FIG. 7.
- (10) FIG. 9 is a side perspective view of the interior elements of the reel portion, showing the stopping mechanism and springs.
- (11) FIG. 10 is a schematic exploded view of the main elements of the present invention.
- (12) FIG. 11 is an enlarged schematic view showing a section of the stopping mechanism being engaged by two key elements.
- (13) FIG. 12 is a schematic view as seen from the bottom of FIG. 11.
- (14) FIG. 13 is a perspective view of the interior portion of an embodiment of the leash of FIG. 1, showing the main operating parts and the reels.
- (15) FIG. 14 is a perspective view of certain components of the interior of the reel of FIG. 13.
- (16) In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

(17) Reference will now be made in detail to representative embodiments illustrated in the accompanying drawings. References to “one embodiment,” “an embodiment,” “an exemplary embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such a feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

(18) FIG. 1 is a perspective view of a dual retractable pet leash **100** which has the ability to stop unreeling of each one of the leashes individually as well as simultaneously. The leash **100** has a reel portion **110** and a handle portion **120**. The handle portion **120** has a grip portion **180**, and is connected to the reel portion **110** at a joint **150**. A bulge **190** contains the joint portion shown in more detail in FIGS. 6-12. The handle portion **120** carries two trigger members **130** and **140**, which separately brake the individual leashes. The reel portion **110** is rotatable continuously relative to the handle portion **120** in either rotational direction about the longitudinal axis of the leash **100** containing the connecting portion **150**. Additionally an aperture **702** is shown in the reel portion **110**, where one of the two leashes can exit the reel portion **110**. The advantages and uses of the leash **100** are as discussed further below.

(19) Single retractable leash devices usually work by having a single reel with a single trigger mechanism for controlling the stopping of the reel. Single leash units having two leashes for walking two pets at the same time can be used to restrain multiple pets. When pets pull on their leash, unwinding of a reel occurs; however sometimes the leash must be stopped from unwinding further, for example to prevent a pet from running into traffic or to prevent the pet from approaching another animal or human. When two leashes are used in single leash mechanism, both leashes are subject to tangling with one another due to movement of the pets or of the owner, or both. Additionally, the leashes must be capable of being stopped, both for the safety of the pets and control of the pets in general.

(20) The leash **100** according to the present disclosure provides a twin pet leash that prevents

tangling by permitting the reel portion **110** to rotate relative to the handle portion **120**, and features individually controllable leash stop mechanisms (shown in FIGS. **6-12**) actuated by trigger members **130** and **140**. Even while an individual leash is stopped, the reel portion **110** can rotate relative to the handle portion **120** so that tangling of the pet leashes is avoided. The leash handle portion **120** has the two triggers **130** and **140** that are preferably color coded to match the color of the corresponding leash that it controls, so that a user can readily identify which of the two leashes are to be stopped. Either or both of the leashes can be stopped or released, without interfering with the operation of the other one of the leashes.

(21) FIG. **2** is a perspective view from the side of the pet leash **100** of FIG. **1**. The elements shown in the figure are as described hereinabove with regard to FIG. **1**.

(22) FIG. **3** is a simplified schematic view showing the relative rotation of the two main parts of the leash of FIG. **1**, namely the reel portion **110** and the handle portion **120** with the two main parts **110** and **120** being shown at a ninety degree angle to each other. A double-headed arrow **160** shows the rotational direction for rotary movement of the reel portion **110**. The handle portion **120** can rotate independently as well in the rotational direction shown by the double headed arrow **170**. The reel portion **110** is shown with two leashes **882** and **883** extending therefrom. It will be understood that FIG. **3** is merely schematic and greatly simplified for purposes of clarity. It is noted that the reel portion **110** can rotate continuously, with no limit on the number of rotations, without disrupting the ability of the trigger members **130** and **140** to stop the individual leashes; this is shown with respect to FIGS. **6-12** discussed hereunder.

(23) FIG. **4** is a perspective view of the leash **100** of FIG. **1** showing the trigger elements **130** and **140**, and the handle portion **120** more clearly, and also showing the contours of the other parts. The elements shown in the figure are as described hereinabove with regard to FIG. **1**. Here, a flat surface **114** and a curved periphery **112** of the reel portion **110** is indicated, as well as a flat top wall **112** and a side wall **124** of the handle portion **120**. A side wall **192** of the handle portion **120** carries the trigger portions **130** and **140**.

(24) FIG. **5** is an end view of the leash of FIG. **1**, showing the trigger portion **192** wherein the handle portion **120** is disposed in an uppermost position and the reel portion **110** is lowermost. The elements shown in the figure are as described hereinabove with regard to FIG. **1**.

(25) FIG. **6** is a perspective view of the interior portion of the leash **100** of FIG. **1**, showing the main operating parts and the reels, which are described in detail as follows. The trigger **130** has a cogged arm **132** which engages with the teeth of a pivoting actuator **300** which pivots about a post **340**.

(26) A support wall **320** is shown which supports the elements shown mounted thereon as described hereunder. The support wall **320** has similar elements mounted on the other side thereof for use with the trigger member **140** (not shown in FIG. **6**). The actuator **300** has an end with a projecting pin **301** which engages a slot formed in a brake actuator **360**. The actuator **360** is pivotably mounted about a pin **370** to cause movement of an actuator arm portion **362** into and out of engagement with an outer, annularly-shaped outer member **503** of a braking mechanism **500**. Another actuator **400** is similarly mounted by similar elements (not shown) on the other side of the support wall **320**, the actuator **400** having an extended arm portion **422**, engages a central, cylindrically-shaped member **502** of the braking mechanism **500**. The actuators **400** and **360** cause linear movement respectively of the elements **502** and **503** which cause braking of the reels in the manner discussed hereunder with respect to FIGS. **7-12** below.

(27) The braking mechanism **500** is rotatably secured within a projecting portion **215** having grooves **216** for mounting the mechanism **500** for rotation with the reel portion **110**. A bearing **600** is provided for smooth rotation relative to the handle portion **120**. The cylindrical member **502** has a distal end **508** which selectively engages notches formed in a reel **200**.

(28) As seen in FIG. **6**, the reel **200** is rotatable about a central portion **119** which in an assembled condition will be filled by a pin or post member to support rotation of the reel **200** thereabout. The

annular portion **503** has a distal end **510** that stops the second reel **290** (shown in FIG. 7). In FIG. 6, the distal portions **508** and **510** are shown in the non-stopped condition. Also, the leash opening **702** is shown as well as a second leash opening **704**, so that two separate leashes can extend from the reel portion **110**.

(29) It should be understood that, while the braking mechanism rotates with the reel portion **110**, the actuator arms **360** and **400** do not rotate with the braking mechanism **500**. Instead, they are positioned so that they simply slide relative to the surfaces with which they are engaged, without interfering with one another, as shown in FIG. 12 hereunder.

(30) FIG. 7 is a schematic perspective view showing only the reels **200** and **290**, and stopping mechanism **500**. Here, the portion of the stopping mechanism which engages directly with the reel portions are the distal ends **508** and **510**, as discussed hereinabove with regard to FIG. 6. A spring element **519** is shown which biases one of the members **502** and **503** into a non-braking position; this is shown in further detail with regard to FIG. 9 discussed below.

(31) FIG. 8 is a schematic perspective view of the reels **200** and **290**, and stopping mechanism **500** of FIG. 7, as viewed from below FIG. 7. The elements shown in this figure are as described hereinabove with regard to FIG. 7. The distal portion **508** engages notches in the reel **200** to stop it when moved toward the reel **200** and into engagement therewith. Similarly, the distal portion **510** engages notches in the reel **290** to stop it when moved toward the reel **290** and into engagement therewith.

(32) FIG. 9 is a side schematic perspective view of the interior elements of the reel portion **110**, showing the stopping mechanism **500** and the spring **503** biasing the central member **502** to a non-engaged position, and another spring **505** biasing the outer cylindrical member **503** to a non-engaged position. The reels **200** and **290** are shown.

(33) FIG. 10 is a schematic exploded view of the main elements of the present invention. The reel portion **110** is shown having an extending portion **215** which has ribs **216** and a bearing **600** thereon. More than one bearing can be used, and more ribs could be provided; further, the bearing member can be omitted although this may increase the frictional resistance against rotation between the parts. The braking mechanism **500** is shown, and it fits partially within a recess **121** formed in the handle portion **120** and partially within the interior of the extending portion **215**. The ribs **216** and bearing **600** fit within recessed grooves (shown in dashed outline in FIG. 10) in the handle portion **120**.

(34) FIG. 11 is an enlarged schematic view showing a section of the stopping mechanism **500** being engaged by two key elements **362** and **400**. The elements are as discussed hereinabove, and are shown in this simplified diagram for the sake of clarity. The distal portions of the braking mechanism have been omitted for the sake of clarity.

(35) FIG. 12 is a schematic view as seen from the bottom of FIG. 11. This shows the central cylindrical portion **502** and the outer annular portion **503**. The dashed outlined elements **362** and **400** show that these elements do not interfere with each other during relative rotation with respect to the members **502** and **503**, and the elements **362** and **400** are in sliding engagement therewith regardless of the rotational direction of motion of the braking mechanism **500**.

(36) FIG. 13 is a perspective view of the interior portion of an embodiment of leash **100**. This embodiment differs from the embodiment discussed above in the way in which actuators **300** and **400** contact outer member **503** and cylindrically-shaped member **502**, respectively. Actuators **300** and **400** are pivotably mounted and actuated by triggers **130** and **140** as discussed above. In this embodiment, actuator arm portion **362** is configured to mount a ball bearing **363** such that ball bearing **363** is brought into contact with outer member **503** when actuator **300** is pivoted. This reduces the friction caused by contact between actuator **300** and outer member **503**, which improves the rotational freedom of outer member **503**, and therefore rotation of reel **110**, when braking is engaged.

(37) As seen in FIGS. 13 and 14, the end of actuator arm **362** is shaped to pass through the bore of

ball bearing **363** to retain ball bearing **363**. In some embodiments, actuator arm **362** can be sized to form a press fit with the bore of ball bearing **363**. In other embodiments, ball bearing **363** may be fixed to actuator arm **362** by one or more other techniques, such as adhesives, plastic welding, or mechanical fastening.

(38) In some embodiments, as shown in FIG. **13**, actuator arm **362** is shaped to orient ball bearing **363** such that the rotational axis of ball bearing **363** is approximately parallel to a radius of outer member **503** when ball bearing **363** contacts the surface of outer member **503**. This means that the flat outer surface **364** of the outer ring of ball bearing **363** will be approximately parallel to the surface of outer member **503** on contact. Further, the rotation of outer member **503** and rotation of ball bearing **363** may be aligned to minimize any friction. This means that the linear contact between bearing **363** and outer member **503** is aligned with the corresponding radius of outer member **503**. It should be understood that the pivoting motion of actuator arm **300** means that bearing **363** will not be aligned in this orientation when actuator arm **300** is retracted to its non-braking position.

(39) In other embodiments, actuator arm **362** can orient ball bearing **363** such that outer surface **364** of ball bearing **363** contacts the surface of outer member **503** at an angle. This may be desirable because it can simplify the geometry of actuator arm **362** with respect to accommodating the pivoting motion of actuator arm **300**. In these embodiments the rotation of ball bearing **363** and outer member **502** are still aligned as discussed above to minimize friction.

(40) In other embodiments, outer surface **364** of ball bearing **363** may be a curved surface designed to minimize surface contact between ball bearing **363** and outer member **503**. For example, outer surface **364** can be a concave surface with a curve that makes the inner portion of outer surface **364** further way from the bore of ball bearing **363** than the outer portions of outer surface **364**. This configuration may reduce friction by reducing the contact area between ball bearing **363** and outer member **503**. It also can reduce the importance of precise alignment between outer surface **364** and outer member **503** because the contact area will remain relatively consistent through a certain range of angular contact between ball bearing **363** and outer member **503**.

(41) In the embodiment of FIG. **13**, actuator **400** is the same as the actuator **400** discussed in the embodiments of FIGS. **1-12**. In other embodiments, actuator **400** can also be formed to carry a bearing, as discussed above with respect to actuator **300** in FIG. **13**. In other embodiments actuator **400** can be shaped is configured such that arm portion **422** creates a point contact with the surface of cylindrically-shaped member **502**. Point contact here is intended to refer to a very small region of contact between arm portion **422** and cylindrically-shaped member **502**. This type of contact can reduce the friction between arm portion **422** and cylindrically-shaped member **502** when the brake is engaged, which improves rotation of cylindrically-shaped member **502** and reel **110**. This point contact can be accomplished by tapering or reducing the cross-section of arm portion **422** to the desired minimum contact area. For example, the end of arm portion **422** that contacts cylindrically-shaped member **502** can be formed in a hemisphere to minimize the contact area. In some embodiments, the tip of arm portion **422** can be configured to mount a contact element **423** to further reduce friction and improve wear resistance. For example, contact element **423** can be a spherical ball made out of a hard-wearing material such as a metal. In these embodiments, contact element **423** can be secured to arm portion **422** through any suitable technique, such as via a press fit or through the use of adhesives.

(42) In these embodiments, arm portion **422** is designed to contact the center of the surface of cylindrically-shaped member **502**. This geometry can reduce the importance of rotationally aligning the end of arm portion **422** with the surface of cylindrically-shaped member **502**. Embodiments that include a spherical or hemispherical contact element **423** also assist in this aspect because the curved nature of the contacting surface of arm portion **422** can accommodate different angular alignments between arm portion **422** and cylindrically-shaped member **502**.

(43) In some embodiments, the surface of cylindrically-shaped member **502** may be designed to

improve wear resistance at the contact point with arm portion 422. For example, a wear member 424 can be placed on the surface at the contact point. Wear member 424 may be a thin disc made of wear-resistant material, such as a metal or hard composite. Wear member 424 can be secured to the relevant portion of the surface of cylindrically shaped member 502 by any suitable technique, such as by adhesives. In other embodiments, wear member 424 may be an integral part of cylindrically-shaped member 502, for example as an inset portion of the surface or as a surface coating. In other embodiments, cylindrically-shaped member 502 may be partially or entirely formed from a wear-resistant material, such as a metal, which then eliminates the need for a separate wear member 424.

(44) In other embodiments, the contact between actuators 300 and 400 and their respective members 502 and 503 can use the same technique. For example, both actuators 300 and 400 can use the point contact discussed above with respect to actuator 400, including the wear-resistant techniques as needed. Other embodiments can use a ball bearing for both contact points, as discussed above with respect to actuator 300. In this case actuator 400 would be modified to move the contact point to an outer annular portion of cylindrically-shaped member 502 to reduce contact friction.

(45) While the pins, connections, and support structures shown are exemplary, the present disclosure is not limited thereto. While specific supports are shown, other types can be used, so long as the range of position and the starting and ending positions of the braking mechanism are substantially the same. Also, various suitable resistance elements and spring elements can be used. The support members can be guided in other ways, such as by other types of linkages, or can be connected in other ways to implement the functionality discussed herein. All such variations and embodiments are contemplated as being within the scope of the present disclosure.

(46) It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present invention as contemplated by the inventor(s), and thus, are not intended to limit the present invention and the appended claims in any way. Moreover, the examples described above do not limit the present disclosure to what has been particularly shown and described hereinabove. Rather, the scope of the present disclosure includes both combinations and sub-combinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.

(47) The foregoing description of the specific embodiments will so fully reveal the general nature of the invention that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present invention. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

(48) Various features of the invention which are, for clarity, described in the contexts of separate embodiments may also be provided in combination in a single embodiment. Conversely, various features of the invention which are, for brevity, described in the context of a single embodiment may also be provided separately or in any suitable sub-combination.

(49) The breadth and scope of the present invention should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A pet leash, comprising: a handle configured to be held by a user; a first trigger disposed on the handle and positioned to be actuated by the user; a second trigger disposed on the handle and positioned to be actuated by the user; a reel casing rotatably connected to the handle; a first reel disposed in the reel casing and having a first leash wrapped around the first reel; a second reel disposed in the reel casing and having a second leash wrapped around the second reel; a brake rotatably disposed in the handle and extending into the reel casing, the brake comprising: a first brake member slidably disposed in the brake such that the first brake member can slide into and out of contact with the first reel in the reel casing; and a second brake member slidably disposed in the brake such that the second brake member can slide into and out of contact with the second reel in the reel casing; a first actuator arm pivotably mounted in the handle and linked to the first trigger such that movement of the first trigger pivots the first actuator arm, wherein the first actuator arm is configured to pivot to displace the first brake member towards the first reel; a ball bearing disposed at an end of the first actuator arm and positioned to contact the first brake member when the first actuator arm pivots to displace the first actuator arm; a second actuator arm pivotably mounted in the handle and linked to the second trigger such that movement of the second trigger pivots the second actuator arm, wherein the second actuator arm is configured to pivot to displace the second brake member towards the second reel; a contact member disposed at an end of the second actuator arm and positioned to contact the second brake member when the second actuator arm pivots to displace the first actuator arm.
 2. The pet leash of claim 1, wherein the first actuator arm is configured to position the ball bearing such that an outer surface of the ball bearing is parallel to a surface of first brake member that is in contact with the ball bearing.
 3. The pet leash of claim 1, wherein the first actuator arm is configured to position the ball bearing such that an outer surface of the ball bearing is not parallel to a surface of first brake member that is in contact with the ball bearing.
 4. The pet leash of claim 1, wherein the ball bearing comprises a curved outer surface such that the contact between the first brake member and the ball bearing contacts the curved outer surface.
 5. The pet leash of claim 1, wherein a contact surface of the first brake member that contacts the ball bearing is an annular surface, and wherein the first actuator arm is configured to bring a rotational axis of the ball bearing parallel to a radial direction of the annular surface of the first brake member when the ball bearing contacts the annular surface.
 6. The pet leash of claim 1, wherein the contact member comprises a spherical element fixed to the end of the second actuator arm.
 7. The pet leash of claim 1, further comprising a wear element positioned on the second brake member at the location of contact between the second brake member and the contact member, wherein the wear element is configured to reduce wear of the second brake member caused by the contact member.
 8. The pet leash of claim 1, wherein the wear element comprises a metal disk that is fixed to a surface of the second brake member.
 9. The pet leash of claim 1, wherein the contact member is configured to create a point contact between the second actuator arm and the second brake member.
-